

SWA-MPPI: Socially-Weighted Alignment with Model Predictive Path Integral Control

Cross-Cultural Moral Alignment of LLMs in Logit Space

A training-free, inference-time framework

Paper walkthrough
(NeurIPS-style submission)

Source: `SWA_MPPI_paper/paper_revised.tex`

Technical Deep Dive

Outline

- 1 Motivation & Problem
- 2 Problem Formulation
- 3 Method: SWA-MPPI Pipeline
- 4 Experimental Setup
- 5 Main Results
- 6 Mechanism & Robustness
- 7 Discussion & Limitations
- 8 Conclusion

The cultural-alignment gap in LLMs

- **Moral preferences are culturally contingent.**

The Moral Machine experiment collected **40M+ judgments from 233 countries**, finding systematic cross-cultural variation: East-Asian respondents weight *youth* more; Southern-hemisphere populations show distinct utilitarian calculi; Islamic societies prioritise different social roles. [Awad et al., Nature 2018]

- **LLMs encode a single, predominantly Western moral stance**—inherited from WEIRD-skewed training data. [Henrich et al. 2010; Santurkar et al. 2023]

- **MultiTP** (Jin et al. 2025) operationalises this into country-specific Average Marginal Component Effects (AMCE) over **6 moral dimensions, 107 languages**; finds that *none of 19 contemporary LLMs* achieve broadly satisfactory alignment.

Research question

*Can a frozen LLM be nudged toward country-specific moral preferences at **inference time**, using only what cross-cultural surveys already tell us?*

Why existing approaches don't fit

Approach	Representative work	Limitation for cross-cultural alignment
Fine-tuning / RLHF	Bai et al. 2022; Ouyang et al. 2022	Per-culture labelled data; irreversible weight change
Ad-hoc prompting	Jin et al. 2025	Weak empirical grounding
Activation steering	Arditi et al. 2025; Zou et al. 2023	White-box; per-model engineering
Reward-guided decoding	ARGS (Khanov 2024); Mudgal 2024	Trained reward model unavailable cross-culturally
Culture adapters	CultureManager 2024; CARE-DiO 2024	Per-culture labels; scale poorly >100 countries

None of these are training-free, black-box, country-conditioned, label-free, and weight-preserving simultaneously.

Core insight: *disagreement is signal*

- Within-country moral views are **not monolithic**—age cohorts & social strata exhibit systematically different priorities. [Awad et al. 2018; Kleiman-Weiner et al. 2017]
- The **World Values Survey (Wave 7)** [Haerpfer et al. 2022] provides empirically calibrated within-country variation across 80+ nations.
- **Build WVS-grounded persona agents** per country. When personas *disagree* on a dilemma, that scenario is **culturally contested** → exploit through stochastic optimal control. [Williams et al. 2017 (MPPI)]
- Apply a **bounded, KL-regularised correction** in **logit space** (not generation sampling), with a **Prospect Theory** utility that captures asymmetric gain/loss sensitivity. [Kahneman & Tversky 1979]

Key principle

Self-limiting: when personas agree, the method does *nothing*. The intervention only fires where it matters.

Setting: binary moral choice in MultiTP

- Scenario x presents a **forced binary choice** via decision tokens A (a) and B (b).
- Frozen LLM f_θ : produces logits $z = [z_a, z_b]$, gap $\Delta = z_b - z_a$.
- Spare probability: $p_{\text{spare}} = \sigma(\Delta / T_{\text{dec}})$.
- **Human target**: country AMCE vector $h^{(c)} \in \mathbb{R}^6$ from 40M+ Moral Machine judgments.
- **Model AMCE** (per dimension d):

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{x \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(x)$$

Five binary dims: empirical mean of p_{spare} . Utilitarianism: OLS regression on count difference $|n_{\text{pref}} - n_{\text{non-pref}}|$.

Goal

Find logit correction $\delta^*(x, c)$, *without modifying* θ , so that $\hat{m}^{(c)}$ is closer to $h^{(c)}$ under **Jensen–Shannon divergence** (primary metric).

Four-stage pipeline overview

- 1 **Persona construction** ($N = 4$): three WVS age cohorts + one utilitarian agent.
- 2 **Batched evaluation**: base + N personas in *a single* forward pass $\rightarrow$ per-agent decision gaps δ_i .
- 3 **Adaptive conflict detection**: if inter-agent variance $\nu \geq \tau^{(c)}$, fire MPPI; otherwise return consensus only.
- 4 **PT-weighted MPPI in logit space + positional debiasing**.

Deployment cost

One additional batched forward pass; the MPPI loop runs on **cached scalars** and is **free at the wall-clock level**.

Stage 1 — Persona construction (WVS → natural language)

8 value dimensions extracted from WVS Wave 7 microdata [Haerpfer et al. 2022]:

- Gender egalitarianism (Q58P, Q59P, Q60P); Religiosity (Q6P); Trust (Q43P)
- Moral permissiveness (Q50, Q52P, Q54P); Work centrality (Q5P)
- Family importance (Q1P); Personal autonomy (Q39P); Meritocracy (Q40P)

Procedure: per country, segment respondents into *young* (18–35), *middle* (36–55), *older* (>55); compute per-cohort dimension means; map to natural-language descriptors (e.g. religiosity ratio >0.85 → “deeply religious”).

Persona prompt template:

You are a [age-cohort] adult from [country]. The cultural values typical of your social group are: [WVS statistics]. When making moral judgments, you tend to [WVS-grounded description].

Fourth persona: country-invariant **utilitarian** agent (maximises lives saved).

Why this matters

Personas reflect **measured population structure**, not author stereotypes—mitigating toxicity risks of naive cultural prompting. [Deshpande et al. 2023]

Stage 2 — Batched forward pass & gaps

Single batched forward pass (base + N personas):

$$z_{\text{base}}, z_1, \dots, z_N = f_{\theta}(x; \emptyset, s_1, \dots, s_N).$$

Per-category logit temperatures $T_{\text{cat}} \in \{4.0, 3.5, 1.5\}$ scale logits before computing gaps:

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N} \sum_i \delta_i \text{ (consensus)}, \quad r_i = \delta_i - \delta_{\text{base}} \text{ (reward)}.$$

- δ_i : persona i 's decision gap on this scenario.
- $\bar{\delta}$: collective consensus.
- r_i : how strongly persona i *pulls away from the culturally-neutral baseline*—a large $|r_i|$ indicates a moral direction the vanilla model does not exhibit.

Stage 3 — Adaptive conflict detection $\tau^{(c)}$

$$\tau^{(c)} = \text{Percentile}_{65}\left(\left\{\text{Var}(\delta_1^{(s)}, \dots, \delta_N^{(s)})\right\}_{s=1}^{n_{\text{cal}}}\right), \quad n_{\text{cal}} = 50.$$

$$\text{MPPI fires} \iff \text{Var}(\delta_1, \dots, \delta_N) \geq \tau^{(c)}.$$

- Calibrated on **50 unlabelled scenarios**—uses only inter-agent variance; **never observes human AMCEs**.
- Targets a **35% MPPI trigger rate** (65th percentile).
- Main experiments: 20% held-out for τ , 80% for evaluation.

Robustness evidence (Appendix §A)

Held-out vs. in-distribution calibration over 15 countries: $|\Delta\text{JSD}| = 0.0006$ (*below* the bootstrap CI of ± 0.004); 12/15 countries $A \leq B$. $\rightarrow \tau$ does *not* overfit.

Stage 4a — Prospect Theory utility (per agent)

Cooperative decomposition with $\lambda_{\text{coop}} = 0.7$:

$$U_i(\epsilon_k) = (1 - \lambda_{\text{coop}}) v(r_i \tilde{\delta}_k) + \lambda_{\text{coop}} v(\bar{r}_{-i} \tilde{\delta}_k), \quad \tilde{\delta}_k = \bar{\delta} + \epsilon_k.$$

Classical Kahneman–Tversky value function [Kahneman & Tversky 1979]:

$$v(x) = \begin{cases} x^\alpha & x \geq 0 \\ -\kappa |x|^\beta & x < 0 \end{cases} \quad (\alpha = \beta = 0.88, \kappa = 2.25).$$

- Sign of $r_i \tilde{\delta}_k$: *positive* if perturbation moves *with* persona i , *negative* otherwise.
- $\kappa = 2.25 \rightarrow$ losses are weighted $\approx 2.25\times$ more than gains $\rightarrow$ **loss-averse alignment**.
- Avoids committing to a single δ_{target} , which the multi-persona setup is designed to avoid.

Stage 4b — Collective objective + MPPI weights

Free-energy collective utility:

$$U_{\text{total}}(\epsilon_k) = \frac{1}{N} \sum_{i=1}^N U_i(\epsilon_k) - \alpha_{\text{KL}} \frac{\epsilon_k^2}{2\sigma^2}, \quad \alpha_{\text{KL}} = 0.05.$$

Importance-weighted MPPI update with $K = 128$ samples:

$$w_k = \frac{\exp(U_{\text{total}}(\epsilon_k)/\beta)}{\sum_{k'} \exp(U_{\text{total}}(\epsilon_{k'})/\beta)}, \quad \delta^* = \sum_{k=1}^K w_k \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, \sigma^2).$$

Key design choice

$\beta = T_{\text{dec}} = 0.5$: aligns MPPI smoothing scale with the final sigmoid decision temperature. Effective sample size $K_{\text{eff}} = 1 / \sum_k w_k^2 \approx 15\text{--}40$ out of 128 $\rightarrow$ *moderate concentration* (balances exploration with exploitation).

Stage 4c — Positional debiasing & final prediction

- **Recency / position bias** is a documented LLM failure mode. [Liu et al. 2024 “Lost in the Middle”]
- Each scenario is evaluated **twice**:
 - original ordering, and
 - with **both** choice labels ($A \leftrightarrow B$) and group labels (Group A $\leftrightarrow$ Group B) swapped *in the native script*.

- Debaised gap:

$$\delta_i = \frac{1}{2} (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})}).$$

- Final prediction:

$$p_{\text{spare}} = \sigma\left((\bar{\delta} + \delta^*) / T_{\text{dec}}\right), \quad T_{\text{dec}} = 0.5.$$

Hyperparameters (all values fixed in advance)

Parameter	Symbol	Value	Source
Persona agents	N	4	Design
MPPI samples	K	128	Design
Perturbation std.	σ	0.3	Validation
Cooperation weight	λ_{coop}	0.7	Validation
KL penalty	α_{KL}	0.05	Validation
PT curvature	$\alpha = \beta$	0.88	Kahneman & Tversky 1979
PT loss aversion	κ	2.25	Kahneman & Tversky 1979
Decision temp.	T_{dec}	0.5	Validation
Trigger rate	ρ_{target}	0.35	Design
T_{cat} (Species/Gender/Others)		4.0 / 3.5 / 1.5	Category

Benchmark, languages, preprocessing

- **MultiTP** [Jin et al. 2025]: trolley vignettes, native languages, 6 moral dimensions: *Species, Gender, Age, Fitness, Social Value, Utilitarianism*.
- **15 countries, 12 languages**: en, zh, ja, ko, de, fr, pt, ar, vi, hi, ru, es.
- **Native-language evaluation**: framing, characters, lane labels, questions all in target language; only A/B decision tokens remain English (verified single-token in every model vocabulary).
- **Preprocessing**: dedup → quality filter (drop equal-size Utilitarianism) → category cap (max 80) → up-sample (min 36 / dim) → 310–500 scenarios per country.
- Adaptive $\tau^{(c)}$ calibrated on 50 unlabelled scenarios (variance only, no AMCE).

Models, decoding, baselines, metrics

Six open-weight LLMs (single H100 80GB, Unsloth, greedy decoding):

- Qwen2.5-72B-Instruct (Int8) [Qwen team 2025]; Qwen2.5-32B / 7B (4-bit)
- Llama-3.1-70B-Instruct (FP8) [Grattafiori et al. 2024]; Llama-3.1-8B (4-bit)
- Mistral-Large-2407 (4-bit) [Mistral AI 2024]

Baselines on Llama-3.1-70B: Vanilla; Country-Tailored (B1); WVS-Profile (B2); PRISM-style (B3) [Kirk et al. 2024]; Activation Steering [Arditi et al. 2025].

Metrics:

- **JSD** (primary, $\downarrow$): Jensen–Shannon distance between model and human AMCE vectors.
- **Pearson r** ($\uparrow$): correlation across 6D AMCE vector.
- **MAE** ($\downarrow$): mean absolute AMCE error.
- Bootstrap 95% CIs (1000 resamples) $\rightarrow$ noise floor ± 0.004 JSD.

Honest disclosure

All results are **single-seed** (seed 42); $K = 128$ MPPI draws per scenario. Across-seed variance not characterised.

Cross-model summary (read by regime, not by ranking)

Model	Size	Mean JSD ↓		Improv.	Pearson r ↑	
		Vanilla	SWA-MPPI		Van.	SWA
Qwen2.5-72B	72B	.0859±.005	.0536±.003	+37.6%	0.548	0.696
Llama-3.1-70B	70B	.0937±.006	.0612±.004	+34.7%	−.026	0.384
Llama-3.1-8B [‡]	8B	.0533±.004	.0436±.003	+22.6%	−.247	0.271
Qwen2.5-7B [‡]	7B	.0522±.004	.0426±.003	+18.4%	0.252	0.402
Mistral-Large	126B	.0457±.003	.0430±.003	+5.8%	0.658	0.550
Qwen2.5-32B	32B	.0632±.004	.0633±.004	−0.2%	0.625	0.550

[‡] 4-bit quantised: low vanilla JSD partly reflects logit flattening (small denominator). *We deliberately do not aggregate these regimes into a single headline number.*

Reading the four regimes

Regime 1 — Strong 70B-class backbones (main claim)

Qwen2.5-72B and Llama-3.1-70B: vanilla JSD high enough to leave real headroom; logits well-spread; persona disagreement informative. **Mean JSD drops from $\sim 0.09 \rightarrow \sim 0.05$ ($\Delta \approx 0.032$ absolute, -37.6% / -34.7%).** Pearson r moves into a meaningfully positive range. **This is where the main empirical claim lives.**

Regime 2 — Small 4-bit quantised models (cautious read)

Qwen2.5-7B, Llama-3.1-8B: low vanilla JSD partly explained by 4-bit logit flattening toward chance. SWA-MPPI yields directional improvement, but % gain is amplified by a small denominator.

Regime 3 — Mistral-Large (informative negative)

JSD improves only marginally; Pearson r *decreases for 11/15 countries*. Hypothesis: on already-aligned logits, MPPI compresses the AMCE vector toward the population mean, improving distance but degrading rank. **Reported, not buried.**

Regime 4 — Qwen2.5-32B (diagnosed failure)

Logit concentration at decision token: fixed $\sigma = 0.3$ perturbations land outside effective logit range $\rightarrow$ importance weights $\rightarrow$ uniform $\rightarrow$ MPPI degenerates to random noise. Logit entropy $H = 0.521$ nats vs. reference $H_{\text{ref}} = 2.5$. **Model-level diagnostic, not silent failure.**

Baseline comparison on Llama-3.1-70B

Method	Mean JSD $\downarrow$	Pearson r	vs. SWA	$ \Delta $
SWA-MPPI (ours)	.0612 $\pm$.004	0.384	–	–
Country-Tailored (B1)	.0680 $\pm$.004	0.233	+11.1%	.0068
PRISM-Style (B3)	.0699 $\pm$.005	0.148	+12.5%	.0087
WVS Profile (B2)	.0737 $\pm$.005	0.155	+16.9%	.0125
Activation Steering	.0877 $\pm$.006	–0.208	+30.2%	.0265
Vanilla	.0937 $\pm$.006	–0.026	+34.7%	.0325

- SWA-MPPI beats the strongest prompting baseline by an absolute **0.0068 JSD**.
- **Activation steering performs below vanilla** $\rightarrow$ contrastive vectors interfere with moral reasoning on this benchmark.
- All deltas are above the bootstrap noise floor of ± 0.004 .

Per-country results: Qwen2.5-72B (13/15 improved)

Country	Lang.	JSD Van.	JSD SWA	r Van.	r SWA	% Improv.
USA	en	.1066	.0308	0.400	0.822	+71.1
Germany	de	.0603	.0603	0.845	0.747	+0.0
China	zh	.0513	.0371	0.851	0.762	+27.8
Japan	ja	.0986	.0553	0.634	0.786	+43.9
Brazil	pt	.0754	.0703	0.349	0.529	+6.7
Saudi Arabia	ar	.0778	.0796	0.443	0.527	-2.3
Vietnam	vi	.1302	.0783	0.569	0.456	+39.9
France	fr	.0789	.0479	0.617	0.838	+39.4
India	hi	.0600	.0561	0.670	0.739	+6.5
South Korea	ko	.0720	.0513	0.355	0.645	+28.8
Great Britain	en	.1057	.0294	0.419	0.800	+72.2
Russia	ru	.0800	.0629	0.623	0.784	+21.4
Mexico	es	.0760	.0534	0.605	0.545	+29.8
Nigeria	en	.1103	.0649	0.407	0.615	+41.2
Australia	en	.1047	.0258	0.438	0.841	+75.4
Mean		.0859	.0536	0.548	0.696	+37.6

Germany (ceiling case): vanilla $r = 0.845$, SWA holds steady $\rightarrow$ **self-limiting** property: low disagreement $\rightarrow$ trigger rarely fires.

The “Anglosphere paradox”

Observation

The largest JSD improvements cluster in the **three English-speaking countries** (AU, GB, US), reaching **+71–75% on Qwen2.5-72B**. These are exactly the populations one would naively expect to be *best* served by an English-trained model.

- Vanilla JSDs for AU/GB/US are the *highest* of any country evaluated.
- Whatever the strong 70B Qwen/Llama models have learned about “the average moral subject” lies further from the Moral Machine AMCEs of these populations than from Germany, China, etc.
- **We deliberately do not attribute this post-hoc to RLHF**—we have no intervention isolating RLHF as the cause.
- Reported *as an open question deserving a dedicated follow-up*, not as an explained insight.

What is doing the work? (Mechanism analysis)

Two families of stripped-down baselines that share SWA-MPPI's infrastructure (multi-agent forward pass, two-pass debiasing) and differ *only* in the decoding rule:

Family A — Consensus-shift baselines (deterministic shifts on $\bar{\delta}$):

- Best variant (CS-CLAMP): magnitude clamping closes *most* of the gap.
- SWA-MPPI still wins by an absolute ~ 0.003 JSD $\rightarrow$ residual = **Prospect-Theory asymmetry** on top of bounded perturbations.

Family B — Binary-ARGS (reward-guided decoding kernel):

- ARGS-UNIF: uniform weighting $\rightarrow$ matches PersonaConsensus.
- ARGS-WVS (WVS-salience weighting, no PT): *worse than uniform in 12/15 countries*. $\rightarrow$ **WVS weighting alone does not help**.
- ARGS-PT (Prospect-Theory utility kernel): JSD = 0.0485, $r = 0.514$, best/tied in **13/15 countries** $\rightarrow$ recovers SWA-MPPI.

Conclusion

The active ingredient is the **Prospect-Theory gain/loss asymmetry operating on bounded, KL-regularised perturbations**—exactly what the SWA-MPPI free-energy objective prescribes.

Robustness suite (11 tests, all pass)

Claim	Test	Threshold	Observed
τ calibration robust (aggregate)	$ \text{JSD}_A - \text{JSD}_B $	< 0.004	0.0006
τ calibration robust (per country)	$\max \Delta \text{JSD} $ over 15	< 0.010	0.0041 (BRA)
Held-out τ does not lose	countries with $A \leq B$	$\geq 8/15$	12/15
Dataset preprocessing irrelevant	JSD range over 6 configs	< 0.010	0.0034
T_{dec} insensitivity	JSD span (Sweep A)	< 0.010	0.0074
Uniform T_{cat} insensitivity	JSD span (Sweep B)	< 0.010	0.0073
Per-category T_{cat} insensitivity	JSD span (Sweep C)	< 0.010	0.0077
MPPI $>$ best deterministic shift	$\Delta \text{JSD}(\text{CS-Clamp} \rightarrow \text{SWA})$	> 0	0.003
WVS weighting alone insufficient	$\text{ARGS-WVS} \succ \text{ARGS-Unif}$	$< 8/15$	3/15
SWA-MPPI $\equiv$ ARGS-PT (kernel)	ARGS-PT best/tied	$\geq 10/15$	13/15
Qwen2.5-32B failure: logit concentration	H vs. $H_{\text{ref}} = 2.5$	$H < 1$	0.521 nats
Cross-lingual pipeline generalises	end-to-end runs	8/8	8/8

Each test independently probes one assumption; bootstrap CI ± 0.004 JSD.

Ablation study: Qwen2.5-72B, USA

Configuration	JSD ↓	Pearson r ↑	MAE
Full SWA-MPPI	0.0425	0.724	9.77
No-MPPI (consensus only)	0.0493 (+.007)	0.650	11.02
Always-on MPPI	0.0422 (−.000)	0.729	9.74
Utility: quadratic	0.0429 (+.000)	0.725	9.68
Utility: linear	0.0444 (+.002)	0.706	10.12
Without utilitarian ($N=3$)	0.0610 (+.019)	0.441	11.09
No debiasing	0.0476 (+.005)	0.743	9.67

Hierarchy of importance: *utilitarian persona* (+43.5%) > *MPPI* (+15.9%) > *debiasing* (+12.0%) > *Prospect-Theory* (+4.5%). Trigger rate fires 100% on USA (genuine high-disagreement scenarios). MPPI *flip rate* only **6.0%**: framework **modulates confidence rather than reversing decisions**.

Discussion: failure patterns are model-specific

- **Vietnam on Llama-3.1-70B:** vanilla $r = -0.519$, SWA-MPPI $r = -0.135$. Headline JSD improvement is +65.8%, but 6-D rank ordering is *still wrong*. Most plausible reading: correction compresses Vietnam's AMCE vector toward population mean, which happens to lie closer to Vietnamese targets than vanilla. **Not direction-aware alignment in any strong sense.**
- **Mistral-Large:** JSD +5.8% but Pearson r *decreases for 11/15 countries* (mean $\Delta r = -0.108$). Same JSD/ r dissociation phenomenon at panel level.
- **Qwen2.5-32B:** degrades in 7/15 countries—logit concentration confirmed.

Practical implication

SWA-MPPI's effectiveness depends on the base model's logit calibration quality. A **pre-flight diagnostic** (e.g., logit entropy) could predict when to activate the framework.

Limitations (in order of how much they should temper confidence)

- 1 **Hyperparameter validation is the weakest link.** Five hyperparameters tuned on 200 GPT-4-generated dilemmas with author proxy labels. The 50-scenario MultiTP transfer check is too small. Temperature/dataset sweeps in Appendix are the strongest empirical defence.
- 2 **Single seed.** One run per model with one MPPI random draw per scenario; across-seed variance not characterised.
- 3 **Pearson r in 6D is noisy.** Individual r deltas are directional, not statistically separable. JSD is the metric our claims rest on.
- 4 **Mistral-Large rank-order regression.** JSD improves but r decreases on 11/15 countries—no clean fix.
- 5 **PT parameters fixed a priori.** $\alpha = \beta = 0.88$, $\kappa = 2.25$ from Western experiments; not validated per-culture.
- 6 **Utilitarian persona dominance.** The single most important component—may overwhelm country-specific personas on non-numerosity dimensions (open question).
- 7 **No per-dimension AMCE breakdown.** Aggregating over 6 dimensions hides which dimensions improve / hurt.
- 8 **No human evaluation; WVS-to-trolley linkage indirect; coverage:** 15 of 100+ MultiTP countries.

Broader impact & safeguards

- **Risk:** aligning to harmful majority preferences (e.g., discriminatory norms within populations); reifying stereotypes embedded in WVS averages.
- Four explicit safeguards proposed:
 - 1 **Governance oversight:** SWA-MPPI must *not* override safety-critical guardrails; deployment requires domain experts familiar with target culture.
 - 2 **Periodic persona updates:** refresh as newer WVS waves become available (current Wave 7: 2017–2022) to prevent temporal staleness.
 - 3 **Robustness checks:** audit per-dimension AMCE shifts to detect persona-driven entrenchment of harmful norms.
 - 4 **Explicit ethical bounds in the utility:** clip δ^* on sensitive dimensions (Gender, Social Value) by hard cap or steep penalty in U_{total} → formalising deployment-time audits as decoding-time constraints.

Contributions (summary)

- ➊ **First MPPI in logit space** for cross-cultural moral alignment of LLMs—training-free, black-box, country-conditioned, label-free, weight-preserving.
- ➋ **Persona-disagreement gating** of a Prospect-Theory-weighted MPPI correction: intervention only fires when the decision is culturally contested (self-limiting).
- ➌ **Empirical evidence on six open-weight models** (7B–126B); strongest, cleanest gains on 70B-class backbones (**−37.6% / −34.7%** mean JSD).
- ➍ **Mechanism analysis** via consensus-shift + Binary-ARGS baselines: WVS weighting alone insufficient → **Prospect-Theory asymmetry on KL-regularised perturbations** is the load-bearing ingredient.
- ➎ **Robustness suite** (11 tests, all pass): rules out preprocessing, temperature, τ leakage as confounders.
- ➏ **Honest reporting of failure modes**: Mistral-Large rank-order regression, Qwen2.5-32B logit concentration, Anglosphere pattern as open question.

Future directions

- **Adaptive** σ from per-model logit entropy $\rightarrow$ broader compatibility (handles Qwen2.5-32B failure mode).
- **Beyond binary forced-choice**: multi-option / open-ended moral reasoning $\rightarrow$ lift persona-disagreement signal to a distribution over rationales.
- **Per-country / learned PT parameters** (relax fixed $\kappa = 2.25$).
- **Per-dimension mediation analysis** to validate the WVS-to-trolley mapping.
- **Newer WVS waves** $\rightarrow$ longitudinal drift assessment.
- **Disentangle language vs. culture** via multilingual evaluation within single countries.
- **Decoding-time ethical constraints**: formalise the safeguards inside U_{total} .

Key references

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Thank you!

Questions?

SWA_MPPI_paper/paper_revised.tex

Code: [repository]